

AdaQAT: Adaptive Bit-Width Quantization-Aware Training

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Abstract—Large-scale deep neural networks (DNNs) have achieved remarkable success in many application scenarios. However, high computational complexity and energy costs of modern DNNs make their deployment on edge devices challenging. Model quantization is a common approach to deal with deployment constraints, but searching for optimized bit-widths can be challenging. In this work, we present Adaptive Bit-Width Quantization Aware Training (AdaQAT), a learning-based method that automatically optimizes weight and activation signal bit-widths during training for more efficient DNN inference. We use relaxed real-valued bit-widths that are updated using a gradient descent rule, but are otherwise discretized for all quantization operations. The result is a simple and flexible QAT approach for mixed-precision uniform quantization problems. Compared to other methods that are generally designed to be run on a pretrained network, AdaQAT works well in both training from scratch and fine-tuning scenarios. Initial results on the CIFAR-10 and ImageNet datasets using ResNet20 and ResNet18 models, respectively, indicate that our method is competitive with other state-of-the-art mixed-precision quantization approaches.

Index Terms—Neural Network Compression, Quantization Aware Training, Adaptive Bit-Width Optimization

I. INTRODUCTION

Deep Neural Networks (DNN) have achieved remarkable results in recent years in a wide range of domains. While many inference computations are done in the cloud, it is increasingly desirable to deploy trained DNNs to edge devices, such as mobile phones and wearable devices, due to privacy, security, and latency concerns or limitations in communication bandwidth. However, modern DNNs contain at least millions of parameters and require billions of arithmetic operations. Memory and computational costs make deployment on embedded devices difficult, if not infeasible in many cases. To mitigate these issues, various compression techniques have been proposed, such as pruning [1], weight sharing [2], knowledge distillation [3], and quantization [4]. With the emergence of hardware platforms offering better support for low (*e.g.* recent Nvidia GPUs and Google TPUs) and custom precision (*e.g.* FPGA and ASIC solutions) compute, quantization is at the forefront of methods used to increase the efficiency of DNN model inference.

Many models can be uniformly quantized to 8 bits [5] and in some cases to even binary [6] or ternary [7] representations. Various methods [8]–[11] push the compression limit even further, using different bit-widths at the (sub)layer level. However, it is challenging to find efficient mixed-precision configurations that compress a model with minimal impact on accuracy. There are three main families of methods that attempt to optimize bit-width allocation for model compression: *search*, *metric* and *optimization*-based. Search-based methods iteratively explore the bit-width assignment space and are generally costly to use. Metric-based approaches are much faster, but tend to give sub-optimal results. Optimization methods offer good performance at a reasonable cost, but most results of this type tend to suffer from instabilities during the optimization process (*cf.* [9]), especially if starting far (*e.g.* training from scratch) from an optimized configuration.

In the following, we introduce **Adaptive Bit-Width Quantization Aware Training (AdaQAT)**, an optimization-based method for mixed-precision uniform quantization of both weights and activations. Its defining characteristic is the use of relaxed fractional bit-widths that are updated using a gradient descent rule, but are otherwise discretized for all operations (in forward and backward passes). Compared to previous approaches, our initial tests show that AdaQAT is able to produce efficient quantized DNNs that are comparable to the state of the art, in both training from scratch and fine-tuning settings.

II. RELATED WORK

Depending on when quantization is performed (after or during training), there are two main families of methods used in practice. The first, Post-Training Quantization (PTQ), is fast, but can lead to non-negligible loss in accuracy for very small formats [12]. Quantization Aware Training (QAT), while slower, generally leads to better results and should be preferred for extreme quantization problems.

A. Quantization-Aware Training

QAT methods stem from pioneering work on binary neural networks [6], [13]. At its core, a QAT method consists of